NAME:

1a	1b	1ca	2a	2b	2c	Σ

Test 1: Propositional Logic

Time limit: 45 minutes. Total points: 100.

1. Consider the following statements:

   - *If the coffee machine is broken, the PhD student is not writing their dissertation.*
   - *If the supervisor is not abroad, the PhD student is writing their dissertation.*
   - *If the supervisor is abroad, the coffee machine is not broken.*
   - *If the PhD student drinks chai tea, the coffee machine is broken.*

   (a) Formalize these statements as a theory T in the language $\mathbb{P} = \{a, b, c, d\}$, where the propositional variables a, b, c, and d represent, respectively, *"the supervisor is abroad"*, *"the coffee machine is broken"*, *"the PhD student drinks chai tea"*, and *"the PhD student is writing their dissertation"*. *(10 pts)*

   (b) Use the implication graph algorithm to find all models of T. *(20 pts)*

   (c) Prove by resolution that *"the PhD student does not drink tea"* is valid in T. (Draw a resolution tree corresponding to a resolution refutation.) *(20 pts)*

2. Consider the theory $T = \{(p \to q) \to r,\ \neg r \to \neg q,\ \neg(p \wedge \neg r)\}$ in the language $\mathbb{P} = \{p, q, r\}$.

   (a) Use the tableau method to find all models of the theory T. (Construct a tableau from T exactly as defined in the lecture.) *(20 pts)*

   (b) Determine the number of theories (up to equivalence) in the language $\mathbb{P}' = \{p, q, r, s\}$ that are conservative extensions of the theory T. Justify your answer. *(20 pts)*

   (c) Is T an extension of the theory $T' = \{p \leftrightarrow q\}$ in the language $\mathbb{P}' = \{p, q\}$? If so, is it a conservative extension? Justify your answers. *(10 pts)*